

Feature Representation and Unsupervised Clustering: Comparing Raw Pixels and Histogram of Oriented Gradients on CIFAR-10

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Abstract

Feature representation of images is one of the most important ways to extract information from images for statistical modeling. Feature descriptors such as histogram of oriented gradients (HOG) provide a different representation of images from raw pixels. In supervised learning, HOG has established research and often outperforms raw pixel representations. However, in unsupervised learning, HOG is less established. In this paper, we compare and evaluate raw pixels and HOG representations using K-means clustering under reduced dimensions using principal component analysis (PCA). Using the silhouette scores from clustering, we show that raw pixels provide more coherent and separable clusters than HOG through its difference in decomposition. The code is available at https://github.com/Icential/hog_clustering.

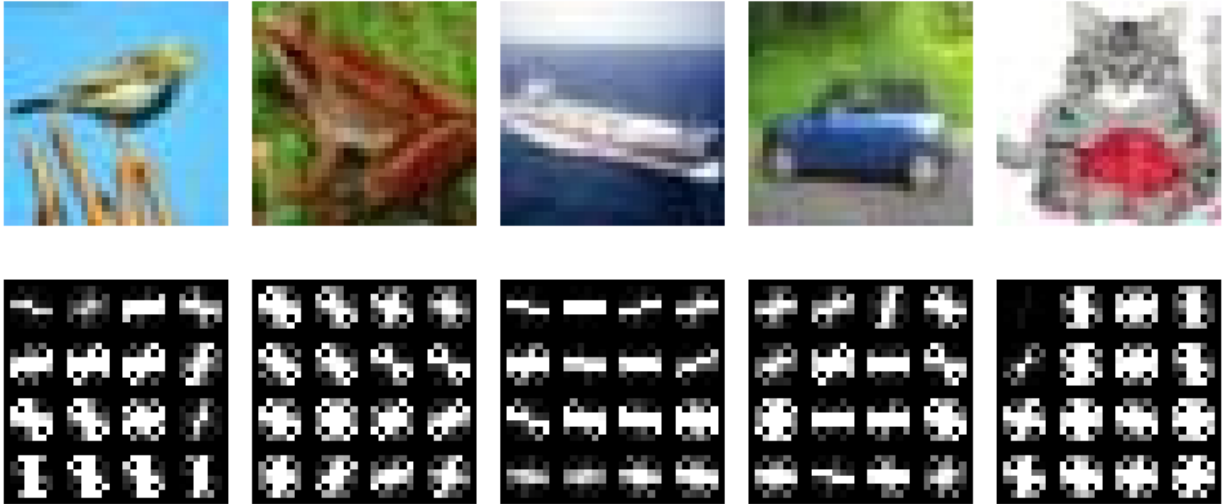


Figure 1: Sample 32×32 RGB images from CIFAR-10 and their corresponding histograms of oriented gradients (HOGs).

1 Background Information and Research Question

In computer vision, images are seen as high-dimensional objects which are often hard to interpret when fed into statistical learning models due to its high dimension. To remedy this issue, other representations of images called feature descriptors are used as a substitute instead of raw pixels. A prominent example of a feature descriptor is the histogram of oriented gradients (HOG) [1]. HOG represents the image by patching it into disjoint regions and creates gradients from each pixel that point towards the direction of greatest pixel intensity change per region (see Appendix A for further explanation).

Previous research on raw pixels and HOG for statistical modeling has been done through supervised learning such as face detection [2] and object detection [3], where HOG is shown to have more robust representations of the images. However, there has been no established research on whether either representation has more coherent semantic meaning in the context of unsupervised learning. To showcase this, we use CIFAR-10 [4], a dataset consisting of 60,000 32×32 pixel RGB images of 10 common objects, to evaluate the differences between raw pixels and HOG representations (see Figure 1 for sample images).

In this paper, we provide an answer to the fol-

lowing research question: Does HOG offer more semantically meaningful representations of images than raw pixels in unsupervised clustering?

2 Method

The test set of CIFAR-10, which consists of 10,000 images, was chosen since it is a well-balanced representation of all the 10 categories in the dataset and avoids computational complexities. For raw pixels, since the images are in RGB, it must first be converted from RGB to grayscale, which is a weighted sum of the three channels into one channel. Then, the 32×32 images are flattened into vectors of $32 \times 32 = 1024$ dimensions. For HOG, we simply input the RGB images into the HOG algorithm as described in [1] using 12 patches and 8 orientations. As clustering is prone to the curse of dimensionality, a high dimension can severely impact how metrics are measured [5]. Thus, we use principal component analysis (PCA) [6] to reduce the high-dimensional representations into lower dimensional representations using the first p principal components, where $p \ll 1024$ (see Appendix B for a visual representation of how increasing raw pixel principal components reconstructs images).

Given the p -dimensional PCA scores of each observation, we use K-means clustering [7] as the method for unsupervised learning to explore se-

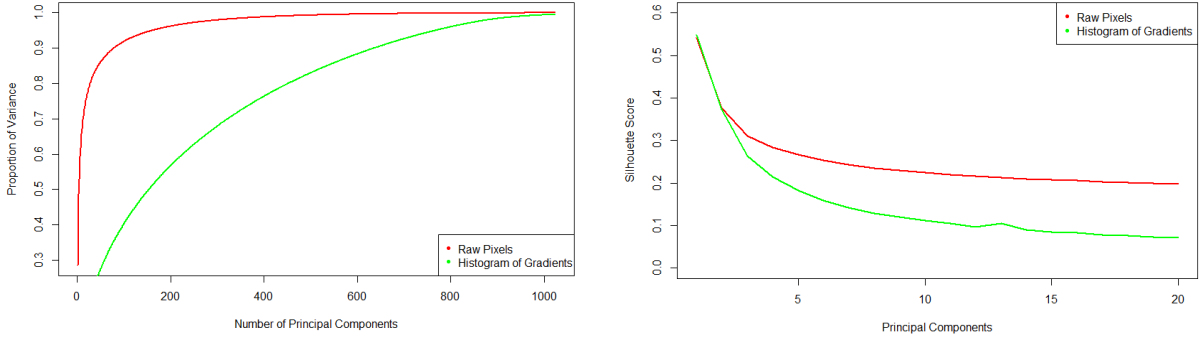


Figure 2: Proportion of variance explained by cumulative principal components (left), and the cluster’s mean silhouette score as a function of the number of principal components used (p) (right).

manically meaningful separations of the images in both representations. For each cluster, we use 10 different random initial centers to ensure the best clustering can be achieved from different trials. Each clustering with n observations is evaluated using the mean silhouette score [8],

$$\bar{s} = \frac{1}{n} \sum_i \frac{b_i - a_i}{\max(a_i, b_i)},$$

where a_i is the intra-cluster distance of observation i , which is the average distance between observation i and other observations in the same cluster. Moreover, b_i is the nearest-cluster distance of observation i , which is the average distance between observation i and other observations from the nearest cluster.

We chose the mean silhouette score $\bar{s}$ as the metric for clustering as it measures the quality of the cluster through its *cohesion* and *separation*. Cohesion is interpreted as how closely related a data point is to other data points within its own cluster. On the other hand, separation is interpreted as how distinct a data point is from data points within its nearest neighboring cluster. We use a grid search iteration approach to determine how many $k \in \{2, 3, \dots, 10\}$ clusters provide the best silhouette score for each p . Also, for both the raw pixels and HOG, we iterate the process by increasing p , which is equivalent to adding more dimensional complexity.

To visually demonstrate the cohesion and separation of clusters, we use uniform manifold approximation and projection (UMAP) [9], a

dimension reduction technique for visualizing high-dimensional data while preserving local and global structure of data points and clusters. It reduces dimensions by constructing a weighted graph of the data and iteratively optimizes a matching low-dimensional layout that resembles the original high-dimensional graph as closely as possible. Here, UMAP is used to reduce the p -dimensional clustering into a 2D scatterplot.

3 Results

Figure 2 shows quantitative results on the different effects of increasing dimensional complexity by increasing the number principal components used (p) for both the raw pixels and HOG representations. While raw pixels exponentially capture the variances of the images, HOG much more slowly and approximately linearly captures the variances of the images. This implies that HOG has much less redundant information spread across its principal components because the first few principal components are unable to capture a significant amount of the variance of the images. On the other hand, for raw pixels, more than 90% of the variance is captured using the first 100 principal components. Therefore, low-dimensional representations of HOG poorly capture information from the original image. Moreover, it becomes increasingly difficult to capture the information as p increases due to the curse of dimensionality.

For clustering a small dimensional representation of the raw pixels and HOG, Figure 2 shows

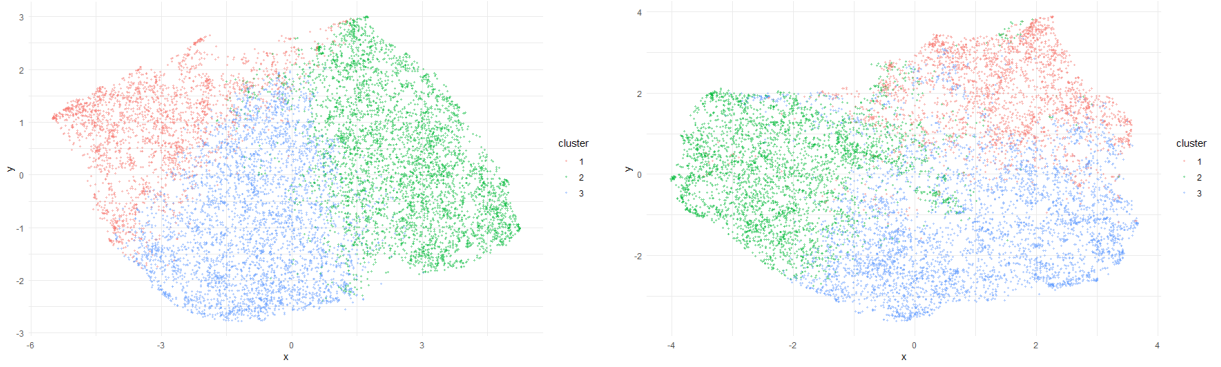


Figure 3: UMAP of K-means clustering at $p = 10$ principal components for raw pixels (left) and HOG (right).

that there is a smooth decaying trend for the cluster’s mean silhouette score as we add more dimensional complexity. While there is only a trivial difference in the mean silhouette scores between these two representations when the first three principal components are used, the decay is worse for HOG as it nears the silhouette score of 0, meaning that the clusters are unable to effectively capture any semantically coherent difference between each other. This aligns with the proportion of variance explained as HOG captures the information of the images significantly more slowly than raw pixels, making the clusters from HOG’s principal components less coherent. Additionally, as increasing p elevates the dimensional complexity of the clustering, HOG is unable to use its high-dimensional and more accurate representation of its principal components due to the curse of dimensionality. Hence, HOG has a bad representation in both low and high dimensions while raw pixels have a more stable representation in low dimensions as indicated by its better mean silhouette score than HOG.

Figure 3 shows qualitative results on the clustering using UMAP. We chose $p = 10$ and $k = 3$ to illustrate the effects of differing silhouette scores with a low-dimensional p and small number of clusters k for the sake of visual clarity. Examining the boundaries of each cluster reveals that the raw pixels have better cohesion and separability as the clusters are more well-shaped and have less overlap among them, creating clearer boundaries. Conversely, for HOG, the boundaries are less coherent and separable since there is more overlap among them. These observations

well reflect the respective silhouette scores that the raw pixels and HOG achieved.

4 Discussion

In this paper, we applied PCA and K-means clustering to CIFAR-10 images to distinguish whether raw pixel representations or HOG representations of images are more semantically meaningful in unsupervised learning. By clustering their low-dimensional representations, we showed that raw pixels offer a better representation of images than HOG due to better cohesion and separability of the clusters.

While this analysis was performed on a dataset with varying image patterns, 32×32 pixel images are generally considered unrealistic low-resolution images that do not reflect how real-world images work. That may explain the lack of useful information that emerges from HOG. Additionally, the clusters arising from both representations in Figure 3 are not well explained by the CIFAR-10 categories (see Appendix C). Furthermore, most of the best silhouette scores that came from the clustering is at $k = 2$, which can be hard to interpret (see Appendix D).

Future work would include repeating the same analysis using more realistic higher-resolution images. Additionally, comparing feature descriptors to neural network activations may be considered as feature descriptors are often used as a pre-processing tool for neural networks.

References

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Appendix A: Histogram of Oriented Gradients



Figure 4: A sample image and its HOG¹.

HOG represents images as gradients that point in the direction of greatest pixel intensity (see Figure 4). HOG first convolves a grayscale image $I(x, y)$ with the kernel $G_x = [-1, 0, 1]$ and $G_y = [-1, 0, -1]^T$, which represents the horizontal gradients $g_x = I * G_x$ and vertical gradients $g_y = I * G_y$ of the image. Each pixel has its magnitude $m = \sqrt{g_x^2 + g_y^2}$ and orientation $\theta = \arctan\left(\frac{g_y}{g_x}\right)$, representing the strength of the change in pixel intensity and direction of the gradient, respectively. Then, the image gets partitioned arbitrarily based on contiguous non-overlapping subregions of the image called patches. Given an arbitrary histogram of orientations, each patch assigns its gradients based on its residing pixels by an accumulative vote in the orientation histogram. To make it more robust to brightness and contrast, the gradients can be normalized [10].

HOG can be visually interpreted as the general geometry or edges of prominent patterns that appear in the image. That is why the HOG representation of images can be optimal for statistical models in supervised learning as it only extracts local edge directions and shapes. In contrast, raw pixels can be noisy due to varying brightness.

¹https://scikit-image.org/docs/stable/auto_examples/features_detection/plot_hog.html

Appendix B: Reconstruction of CIFAR-10 Images Using PCA

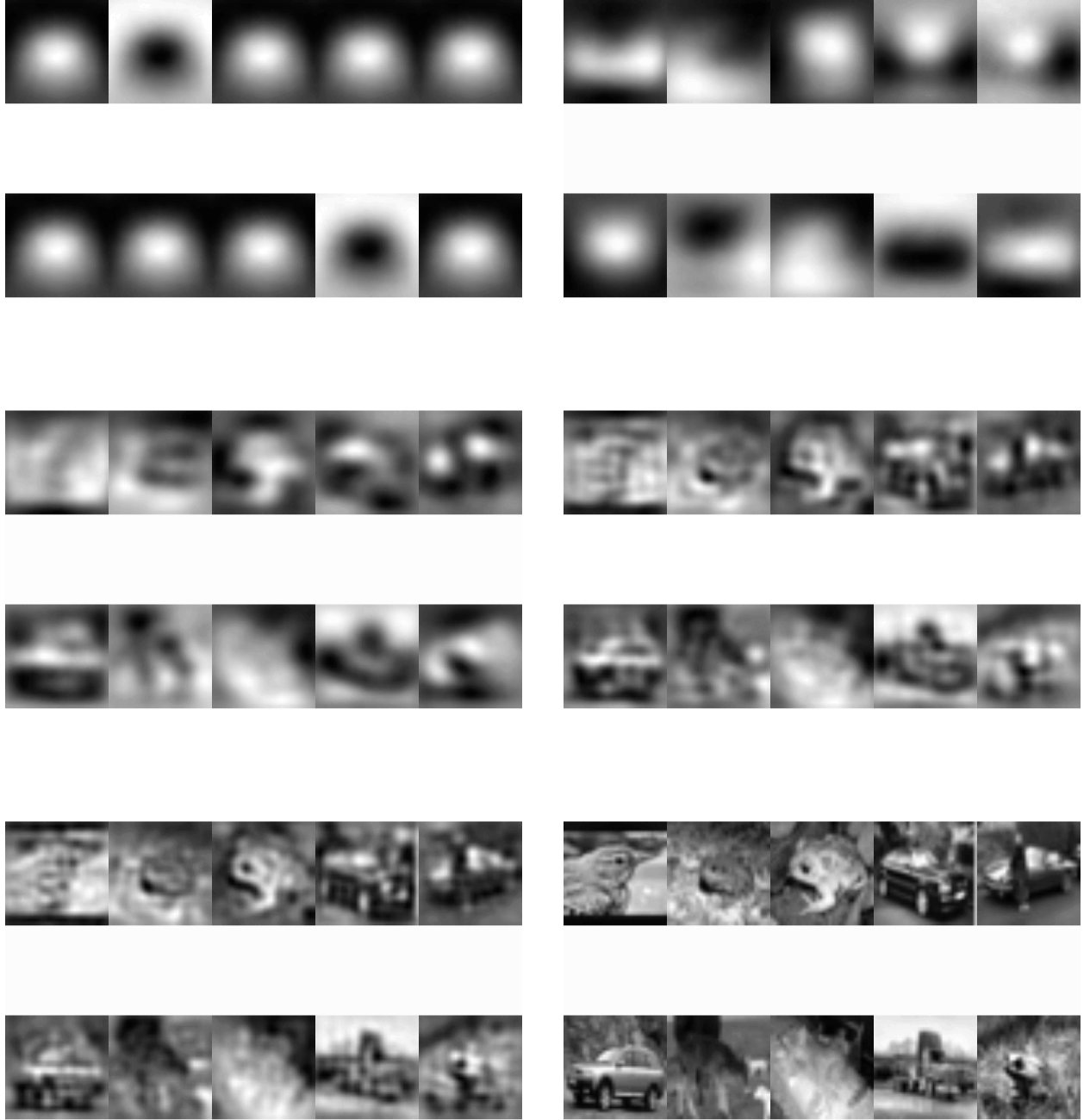


Figure 5: Reconstruction of images from $p = 1, 5, 30, 80, 160, 1024$ raw pixel principal components (left-right, top-bottom). The original image is perfectly reconstructed at $p = 1024$.

As we add more principal components, more detail, shade, lines, patterns and geometry appear. The majority of the first principal components only represents a general shade of the image while latter principal components refine and sharpen the image into its exact patterns (see Figure 5).

Appendix C: UMAP with CIFAR-10 Labels

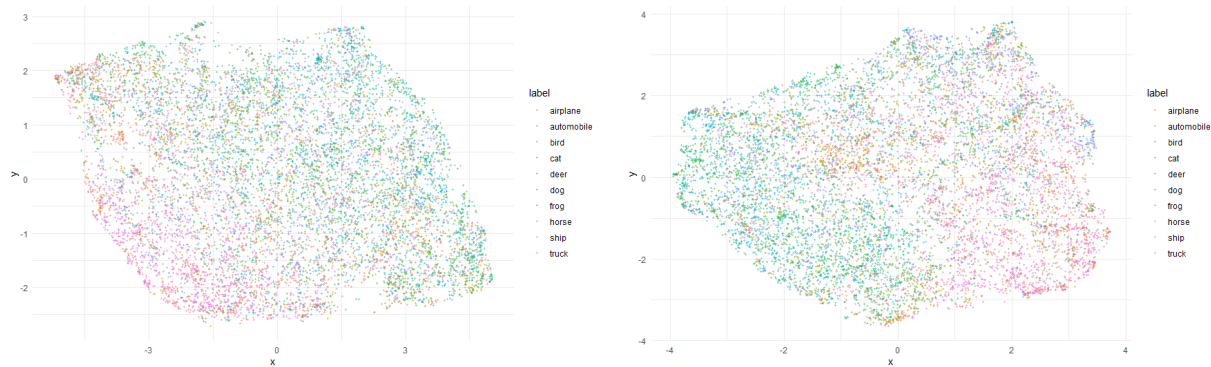


Figure 6: UMAP with color labels from CIFAR-10 categories of K-means clustering for raw pixels (left) and HOG (right).

Appendix D: Optimal Number of Clusters (k)

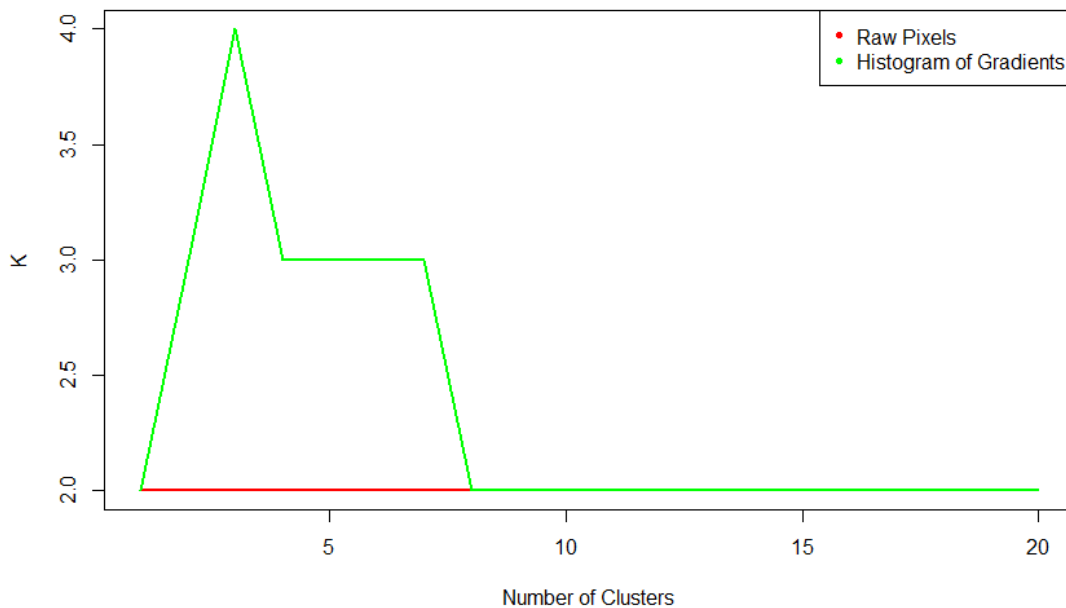


Figure 7: Values of the optimal number of clusters (k) in K-means clustering that resulted in the best silhouette score from Figure 2. A majority of it is at $k = 2$.